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Effects of oscillation amplitude and frequency in narrow-gap laser wire-filling welding of Al alloy 5A06 on the melt pool and the mechanism of pore inhibition

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Outline

[Highlights](#)[Abstract](#)[Graphical abstract](#)[Keywords](#)[1. Introduction](#)[2. Model establishment and instrument](#)[3. Discussion](#)

Highlights

- Circular oscillation enlarges keyhole opening, weakens vapor-jet impact, and markedly suppresses narrow-gap porosity.
- Critical oscillation thresholds are identified for bubble capture and elimination, reducing porosity to negligible levels.
- Narrow-gap confinement intensifies rear-melt vortices; oscillation redistributes flow and prevents collapse-induced bubbles.
- High-frequency (200 Hz) oscillation evens energy input, stabilizes keyhole depth, and enhances bubble escape efficiency.

- A predictive model links oscillation parameters to bubble-capture distance and keyhole behavior for porosity suppression.

Abstract

This study investigates molten pool dynamics and keyhole behavior in narrow-gap oscillating laser wire welding of 5A06 aluminum alloy through numerical simulation and experimental validation. The developed Flow-3D model accurately characterizes molten pool morphology and flow characteristics. Under narrow-gap constraints, conventional laser welding generates strong rear vortices that destabilize the keyhole wall, leading to bubble nucleation. These bubbles become trapped at solidification fronts due to complex melt flow, forming porosity defects. Comparative analysis reveals circular laser oscillation effectively reconstructs molten pool flow patterns by expanding the keyhole opening diameter and reducing the metal vapor jet impact, significantly improving keyhole stability. The study demonstrates critical effects of oscillation parameters (amplitude A , frequency f) on defect control. When $A < 0.6\text{ mm}$, insufficient stabilization results in porosity levels ($\approx 8.2\text{ vol\%}$) comparable to non-oscillating conditions. At $A \geq 1.2\text{ mm}$ and $f \geq 150\text{ Hz}$, periodic keyhole motion actively captures and removes bubbles through the keyhole channel. A porosity suppression threshold model was established based on molten pool flow analysis and bubble trajectory tracking, incorporating maximum capture distance and keyhole dynamic characteristics. This model provides theoretical guidance for process optimization in narrow-gap laser welding applications.

Graphical abstract

4. Conclusion

CRedit authorship contribution statement

Disclosure statement

Declaration of competing interest

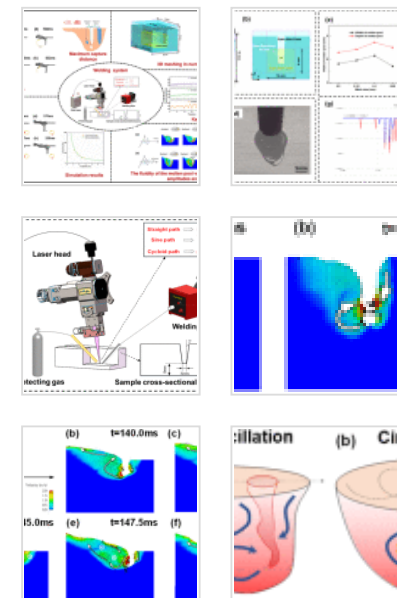
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Data availability

References

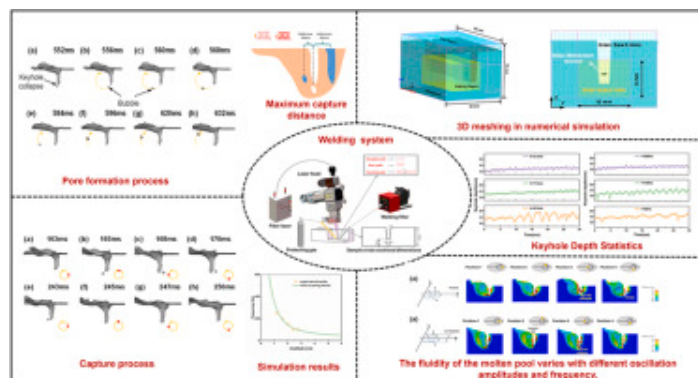
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Keywords

Oscillating laser welding; Narrow-gap aluminum welding; Flow field simulation; Keyhole stability; Porosity inhibition mechanism

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1. Introduction

Al alloys are featured by low specific weight, high specific strength, and excellent thermal conductance, all of which are essential for effective weight reduction of thick plates. Hence, Al alloys are ideal candidates for lightweight applications in transportation and aerospace [1]. In virtue of high energy density, high processing efficiency, low heat input required, and large aspect ratio of weld, laser welding has been widely applied in the welding of Al alloys [2,3] and assembly welding of complex, thick-walled structures [4]. In order to further enhance welding efficiency and relieve thermal stress accumulation caused by multiple thermal inputs, narrow gap welding has been proposed and widely

[Table 1. Heat source parameters of narr...](#)

[Table 2. Physical properties of 5A06 alu...](#)

[Table 3. Average size and fluctuation of ...](#)

applied in the welding of thick plates [5]. With a narrow gap groove, narrow gap welding can significantly reduce filler metal consumption and heat input, thereby enhancing production efficiency and minimizing welding deformation.

Oscillating laser welding has been widely employed for Al alloys due to its capability of suppressing welding defects. Meanwhile, recent FEA-based hybrid machine-learning surrogate-modeling studies have effectively predicted key weld-pool metrics—including penetration depth, pool width, and molten-pool area—further underscoring the strong coupling between thermal behavior and weld-pool morphology in aluminum-alloy laser welding [6]. Liao et al. [7] reported that oscillating laser welding can inhibit pore formation and melt pool collapse in the welding of Cu alloys of medium thickness, indicating that laser oscillation-induced high-frequency stirring and mixing effect could effectively enhance the weld appearance of the fusion zone and facilitate grain refinement. Schultz et al. [8] investigated the Al alloy gap-bridging capability of splice plates using an oscillating laser and found that the oscillating laser could significantly increase the area of the heat-affected zone, and improve the gap-bridging capability of the laser under large gaps and heterogeneous thickness. Circular oscillation has been demonstrated to be an effective way to achieve high-quality weld appearance [9,10], thereby satisfying requirements on penetration depth and defect-free weld. Wu et al. [11] reported that compared with no oscillation and transverse oscillation, circular oscillation led to reduced aspect ratio of weld, homogeneous energy distribution, and minimized welding spatter; the probability of pore formation was reduced by increasing the keyhole size.

Yamazaki et al. [12] demonstrated that amplitude and frequency had significant influences on the weld profile. In cases of high amplitudes, the weld exhibited a non-uniform profile where the penetration depth is greater at both ends and smaller in the middle part. He et al. [13] investigated the effects of an oscillating laser on the dynamic behaviors of the keyhole and melt flow in narrow gap laser-MIG integrated welding of thick Al alloy plates. The results showed that keyhole stability was improved by introducing an oscillating laser, wherein its open size increased by 130% compared with

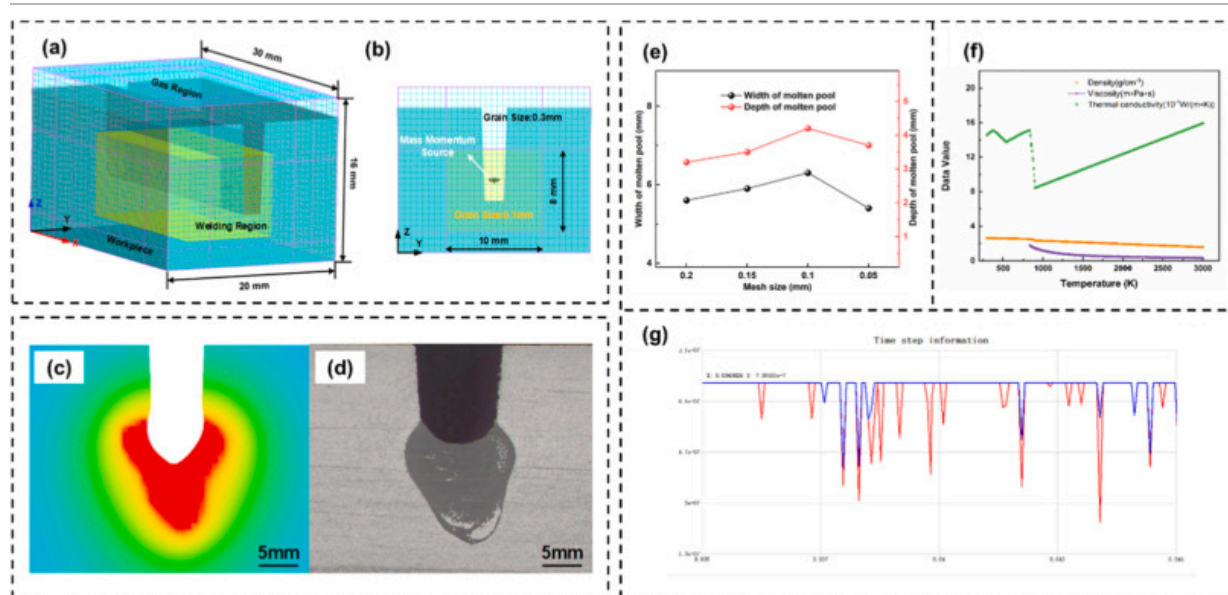
that under a non-oscillating laser; irregular fluctuations in both the keyhole wall and the bulge were effectively suppressed. Chuang et al. [14] investigated the mechanism by which the laser oscillation mode influences welding stability, and reported optimization of weld appearance and pore inhibition. The results showed that circular oscillation effectively inhibited the periodic fluctuations of the keyhole, thereby reducing bubbles in the melt pool and eventually inhibiting pore generation.

Compared with previous studies, this work systematically reveals pore generation and suppression mechanisms in narrow-gap oscillating laser wire-filling welding of aluminum alloys through numerical simulation. Annular laser oscillation is found to enhance keyhole stability by improving rear molten flow, and a dual suppression mechanism—keyhole stabilization and bubble recapture—is proposed. A predictive model for maximum bubble capture distance and amplitude–frequency threshold provides theoretical guidance for complete pore elimination and defect-free welding process optimization.

2. Model establishment and instrument

2.1. Model establishment and validation

Flow-3D was employed to perform the numerical simulations. The welding process started 4mm from the origin along the x-direction, and the computational domain was constructed using an STL file generated from the actual narrow-gap groove geometry. A non-uniform hexahedral mesh was adopted, with grid sizes of 0.3mm in the outer region, 0.1mm in the central welding zone, and 0.1–0.3mm in the transition area, as shown in Fig. 1(a and b). The size ratio between adjacent cells was kept below 1.4 to ensure accuracy, resulting in a total of 367,200 cells.



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Fig. 1. 3D meshing in numerical simulation of narrow gap oscillating laser wire filling welding: (a) Meshing; (b) Grid refinement area; (c) Simulation results; (d) Experimental results; (e) Mesh independence test; (f) Thermo-physical properties of 5A06 aluminium alloy as a function of temperature; (g) Time-step validation.

As seen in Fig. 1(e), mesh independence was evaluated using four different mesh sizes in the central welding zone, with melt pool depth and width as assessment metrics. Considering both accuracy and computational cost, a mesh size of 0.1 mm was selected. All equations were solved using the PISO algorithm, with convergence criteria set to 10^{-6} for the energy equation and 10^{-3} for all other variables. Building upon the mesh configuration and solution strategy described above, the convergence behavior of the numerical simulation was further evaluated. Convergence was assessed by comparing the difference between the stability limit and the time-step size. A larger difference indicates higher residuals, poorer convergence, and consequently longer computational time. Ideally, the time-step size should approach the stability limit to achieve optimal

numerical stability. The simulations demonstrated that a mesh resolution of 0.1 mm provided satisfactory convergence while producing weld bead profiles that closely matched the experimental observations. Therefore, this mesh size was selected for subsequent calculations.

Laser welding involves highly complex thermo-physical phenomena, including laser-sample absorption loss, multi-phase transformations in the sample, movement of the keyhole free surface, and flow behaviors within the melt pool. In order to simplify the numerical simulation (to enhance numerical efficiency without degrading calculation accuracy), the following assumptions were made on the proposed model: (1) the molten metal is an incompressible Newtonian fluid, and the melt is in a laminar flow; (2) during calculations, the dynamic behaviors of the keyhole and melt pool are determined by surface tension and recoil pressure; (3) the welding process is in a quasi-steady-state, and the keyhole formation process can be neglected in the calculation; (4) the metal is isotropic and the variation of its material properties is a function of temperature only, and the influences of other factors such as pressure on physical parameters are not taken into consideration; (5) to better characterize the laser oscillation, the influences of multiple reflections of the laser is neglected, and a rotating Gaussian volumetric heat source was employed as the input of the heat source.

Since the energy required for wire melting must be considered, the filler wire was modeled as a continuous liquid stream with specific initial mass and momentum injected into the melt pool. Herein, phase transition during wire melting and potential constraints exerted by the narrow groove on the rigid wire were neglected, and the wire feeding angle was kept constant. In this study, we focused exclusively on the influences of wall constraints on melt flow characteristics during oscillating laser wire filling welding, while other parameters were selected within their optimal ranges to exclude the influences of shielding gas on flow characteristics. In order to simplify the numerical simulation, the composition of the filler wire was assumed to be similar to the base material, and it was assumed that they had consistent densities.

The inflow of molten wire into the melt pool is modeled as a filling process of molten metal carrying thermal energy, and the heat carried by the molten wire (Q_{wire}) can be determined by:

$$Q_{Wire} = \rho \pi r^2 w_f H_d = \alpha_3 Q \quad (1)$$

Where ρ is the filler density, r_w is the wire radius, and w_f is the wire feeding rate; H_d is the total enthalpy of the molten wire, wherein one portion of the enthalpy is consumed to sustain the liquid state of the wire material and the remainder is transferred to the melt pool as an enthalpy addition [15]:

$$Q_{add} = \rho \pi r_w^2 w_f C_{pl} (T_d - T_l) \quad (2)$$

Where C_{pl} is the specific heat of the liquid wire, T_d is the wire temperature, and T_l is the liquidus temperature of the wire.

Total laser energy can be described by:

$$q_{total}(x, y, z) = q_c(x, y, z) + q_r(x, y, z) + Q_{Wire} \quad (3)$$

$$\alpha_1 + \alpha_2 + \alpha_3 = 1 \quad (4)$$

The model parameters are summarized in [Table 1](#).

Table 1. Heat source parameters of narrow gap laser welding corresponding to different oscillation modes.

Oscillation modes	α_1	α_2	α_3	η_1	η_2	h_c	h_r
Circular oscillation	0.1	0.8	0.1	0.9	0.15	3.1 mm	2.5 mm

These assumptions improve computational efficiency but may affect accuracy. Neglecting turbulence, wire melting phase change, and small pore formation can underestimate local flow and pore risk; assuming material properties depend only on temperature and

ignoring laser reflections and shielding gas may affect heat input and surface flow. The model reliability was validated by using the fusion line comparison method. The results indicated that the circular oscillating laser, by simulation, showed consistent cross-sectional morphology of the melt pool with the practical weld under constant parameters. Such correspondence between simulation and experiment is consistent with previous numerical investigations of conduction-mode laser welding of Al 2024 alloy, where the predicted temperature distribution and weld geometry showed similarly high agreement with experimental observations [16].

During the welding process, several material properties vary with temperature, and these changes in thermal–physical parameters, in turn, affect the heat and mass transfer behavior inside the molten pool. Therefore, the key temperature-dependent properties were calculated using the commercial software JMatPro. Taking the 5A06 aluminum alloy used in this study as an example, the specific heat, viscosity, density, and thermal conductivity were obtained through the software, as shown in Fig. 1(f). Other physical parameters were assumed to be temperature-independent and treated as constants, which are summarized in Table 2.

Table 2. Physical properties of 5A06 aluminum alloy.

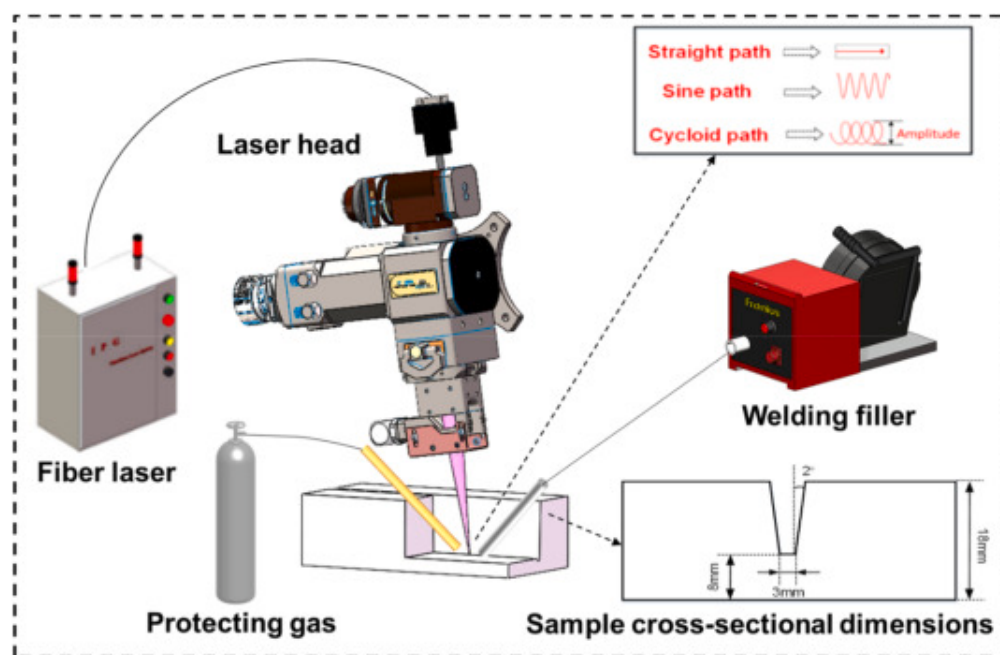
Nomenclature	Symbol	Unit	Value
Solidus temperature	T_S	K	830
Liquidus temperature	T_L	K	940
Specific heat capacity of solid	C_{PS}	$\text{J kg}^{-1} \text{K}^{-1}$	1090
Specific heat capacity of liquid	C_{PL}	$\text{J kg}^{-1} \text{K}^{-1}$	1230
Surface tension coefficient	σ	N m^{-1}	0.914
Temperature coefficient of surface tension	A_σ	$\text{N m}^{-1} \text{K}^{-1}$	−0.00035
Latent heat of fusion	L_m	J kg^{-1}	3.87×10^5

Nomenclature	Symbol	Unit	Value
Latent heat of evaporation	L_v	J kg^{-1}	1.08×10^7
Coefficient of thermal expansion	β	–	1.92×10^{-5}
Convective heat transfer coefficient	h_c	$\text{W K}^{-1} \text{m}^{-2}$	15
Thermal emissivity	ε	–	0.08
Acceleration due to gravity	g	m s^{-2}	9.8
Ambient temperature	T_{ref}	K	300

The differences between experimental and simulation results can be attributed to several factors: (1) during practical welding, the solid wire may experience interference from the wall during feeding into the melt pool, preventing completely stable delivery; (2) the influences of shielding gas on the surface of the melt pool were neglected. This study primarily presents a comparative analysis of the characteristics of the melt pool between circular oscillating laser and non-oscillating laser during narrow-gap oscillating laser wire filling welding.

2.2. Materials and instruments

A push-pull wire feeder (Fronius KD7000) allows wire feed speed adjustment from 0 to 10m/min. The relative positions of the laser head, wire nozzle, and shielding gas tube are illustrated in [Fig. 2](#).



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Fig. 2. Narrow gap oscillating laser wire filling welding system and groove design.

An oscillating laser welding system was used in this study. Fig. 2 shows the oscillating laser welding system used for narrow gap groove wire filling. It comprises a fiber laser (wavelength=1070nm, YLS-30000, IPG) and a welding head (focal length=470mm, spot diameter at 0 defocus=0.6mm, IPG D50 Wobble). The galvanometer module integrated within the laser head enables programming of the laser beam trajectory. As indicated, linear vertical oscillation and circular oscillation were achieved, with a maximum amplitude of 5 mm and a maximum frequency of 600Hz. A push-pull wire feeder (Fronius KD7000) was used, and it allows wire feed speed adjustment from 0 to 10m/min. The relative positions of the laser head, the wire feeder, and the shielding gas tube are illustrated in Fig. 2. The specimen was an 18mm-thick 5A06 Al alloy and processed accordingly to facilitate the study of weld quality in oscillating laser wire filling welding

under narrow gap groove constraints. To ensure the accessibility of the groove bottom by the oscillating laser beam, a groove shown in Fig. 2 was used. Herein, the blunt edge of the groove bottom was 8 mm, the width of the groove bottom was 3 mm, and the single-sided groove angle was 2° . The porosity levels of welds produced under different oscillation frequencies and amplitudes were measured using X-ray radiography (XRD) techniques to validate the reliability of the numerical results.

3. Discussion

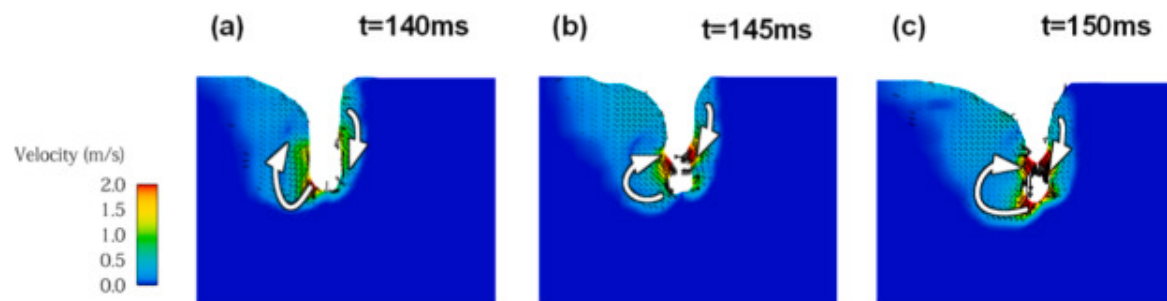
3.1. Dynamic behaviors of the keyhole and melt pool in narrow gap circular oscillating laser welding

3.1.1. Constraining effects of narrow gap on melt flow in circular oscillating laser welding

The narrow gap groove exerts significant constraining effects on the melt pool and considerable influences on the weld appearance. Therefore, the flow behaviors on the longitudinal section of the melt pool under narrow gap constraints were investigated.

Fig. 3 shows the flow behaviors on the longitudinal section of the melt pool in narrow gap oscillating laser wire filling welding with non-oscillation. In this case, the vortex behind the keyhole, directed towards the keyhole wall surface, remains prominent, and its range expands further. At some moments, the upper part of the keyhole is also impacted by the rear vortex, leading to keyhole instability. Similar numerical investigations have likewise shown that, during laser welding of aluminum alloys, the rearward recirculation vortex and steep temperature gradients can induce keyhole instability, making bubbles more prone to being trapped and eventually forming porosity. These findings are consistent with the observations of the present study [17]. In other words, the keyhole stability further deteriorates under the narrow gap constraints and the filling of molten metal into the melt pool. At the forefront of the melt pool, the front wall surface of the keyhole was

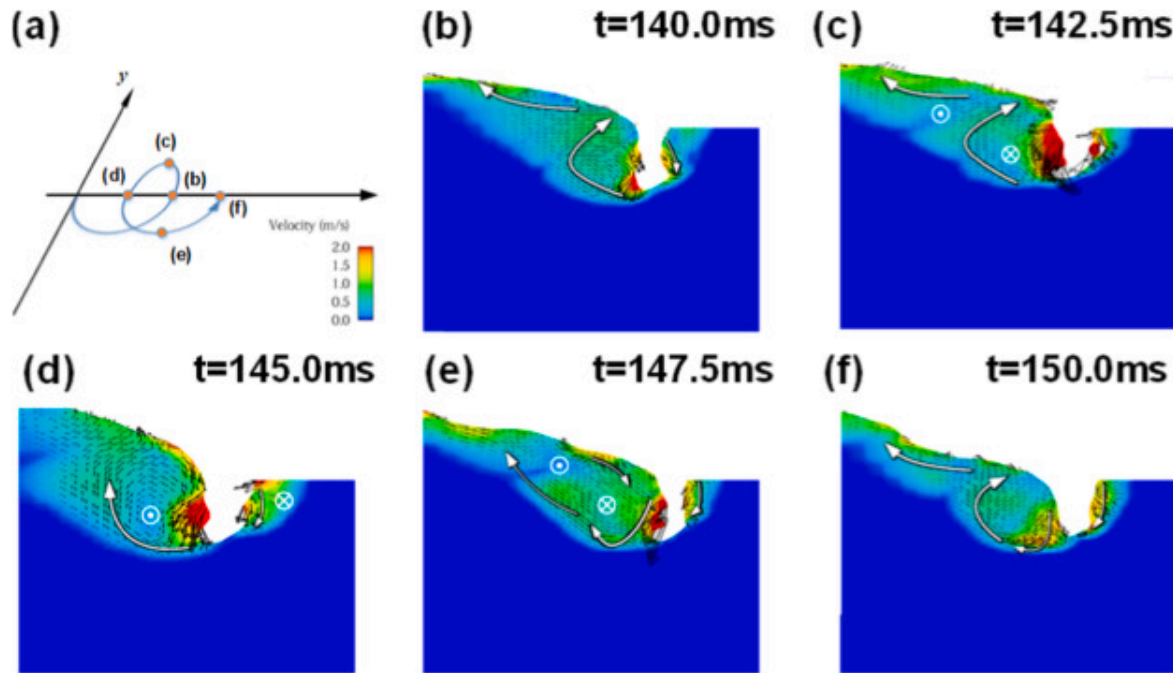
drastically impacted due to the transition behaviors of wire melting. As shown in Fig. 4, under narrow gap constraints, the melt pool was significantly elongated, and the melt accumulation behind the keyhole was significantly greater than that in front of the keyhole, resulting in adverse influences on the keyhole stability.



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Fig. 3. Melt flow patterns of narrow gap oscillating laser wire filling welding with non-oscillation: (a) $t=140.0\text{ms}$; (b) $t=145.0\text{ms}$; (c) $t=150.0\text{ms}$.



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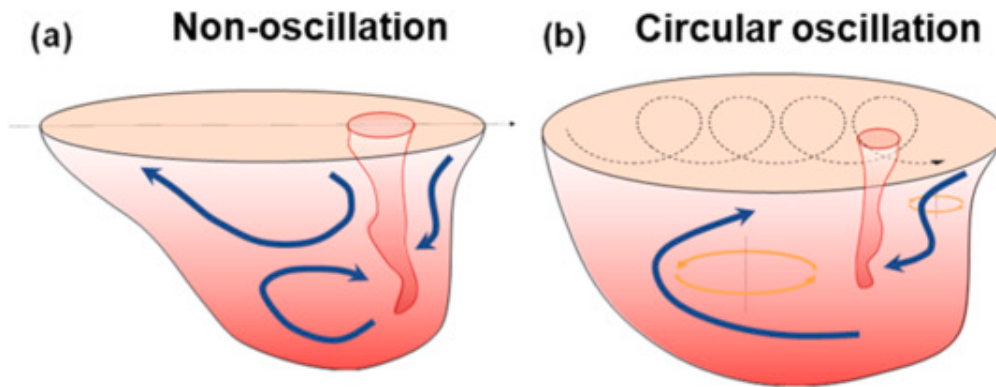
Fig. 4. Flow patterns in the melt pool of narrow gap laser wire filling welding with circular oscillation: (a) Flow field selection position; (b) $t=140.0\text{ms}$; (c) $t=142.5\text{ms}$; (d) $t=145\text{ms}$; (e) $t=147.5\text{ms}$; (f) $t=150.0\text{ms}$.

Fig. 3 illustrates the flow behaviors in the melt pool of circular narrow gap oscillating laser wire filling welding. The unique horizontal vortex induced by the circular oscillating laser remains within the narrow gap; however, its influence on the rearward flow of the melt pool is significantly weakened due to the groove constraints. The vortex behind the keyhole, directed toward the wall surface of the keyhole, can barely be completely suppressed. As a result, the position where the vortex points toward the wall surface shifts upward (targeting the area near the keyhole opening in some cases), resulting in relatively minor influences on the keyhole stability, as shown in Fig. 3(b). Due to the impact of the molten wire at the forefront of the molten pool, the front wall surface of the

keyhole experiences a substantial force, resulting in a significant reduction in the keyhole stability, especially when the keyhole moves to the rearward position of the oscillation circle, as shown in Fig. 3(c).

Compared with narrow gap non-oscillating laser wire filling welding, narrow gap circular oscillating laser wire filling welding showed drastically decreased keyhole depth and melt pool depth, increased open diameter of keyhole, and reduced influences of the flow near the wall surface on keyhole bottom, suggesting that circular oscillating laser can effectively improve melt flow and enhance keyhole stability in this case. This is consistent with previous studies [18].

The effects of circular oscillating laser on melt flow under narrow gap welding are shown in Fig. 5. During non-oscillating laser welding, a significant vortex persists behind the keyhole, impacting the wall surface of the keyhole and causing mechanical instability and bulging. Meanwhile, the molten wire exerts a substantial impact on the forefront of the melt pool under narrow gap constraints, resulting in severe bulging on the front wall surface of the keyhole. Due to both factors, the keyhole stability in narrow gap laser welding degrades drastically, resulting in increased probability of keyhole collapse and bubble formation. A circular oscillating laser can effectively suppress the tendency of the rear vortex directed toward the keyhole wall surface, but cannot completely eliminate the rear vortex. Also, it has negligible inhibitory effects on the impact of the molten metal at the forefront of the melt pool on the keyhole. From the perspective of flow characteristics, the circular oscillating laser can improve keyhole stability and reduce the probability of bubble formation in the narrow gap laser welding. However, the flow direction does not significantly facilitate bubble escape.



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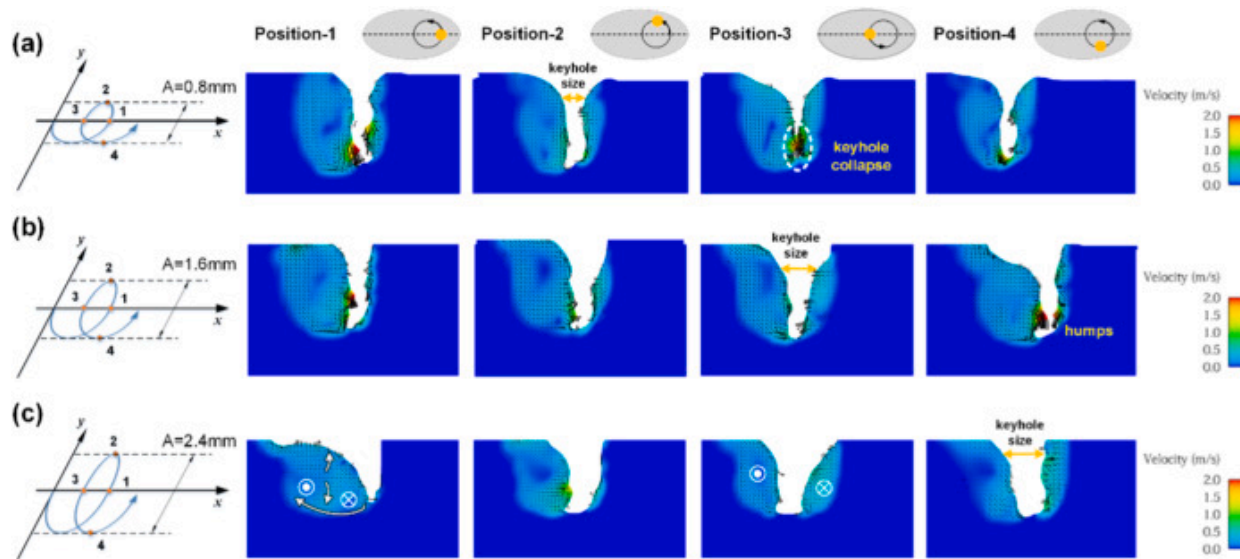
Fig. 5. Flow behaviors in the melt pool of narrow gap laser wire filling welding with (a) No oscillation; (b) Circular oscillation.

Besides oscillation mode, amplitude and frequency also had significant influences on the final welding quality. In this study, the correlation of amplitude and frequency with melt flow was investigated by numerical simulations. Herein, the amplitude is defined as the diameter of circular oscillation of the beam (A , mm), and the frequency is defined as the number of circular motion cycles per unit time (f , Hz).

3.1.2. Effects of amplitude on melt pool and keyhole

The oscillation amplitude of the laser beam directly determines the effective interaction area on the molten pool surface, thereby altering the temperature field and the associated flow behavior. As shown in Fig. 6, when the amplitude is small ($A=0.8$ mm), the flow pattern remains similar to that of linear laser welding. A backward-directed recirculation vortex persists behind the keyhole, and, together with the periodic fluctuation of recoil pressure, weakens the stability of the keyhole wall. In addition, the steep local temperature gradients near the keyhole give rise to strong Marangoni-driven flow

confined to a limited region, indicating that small-amplitude oscillation has only a minor influence on the global flow field.



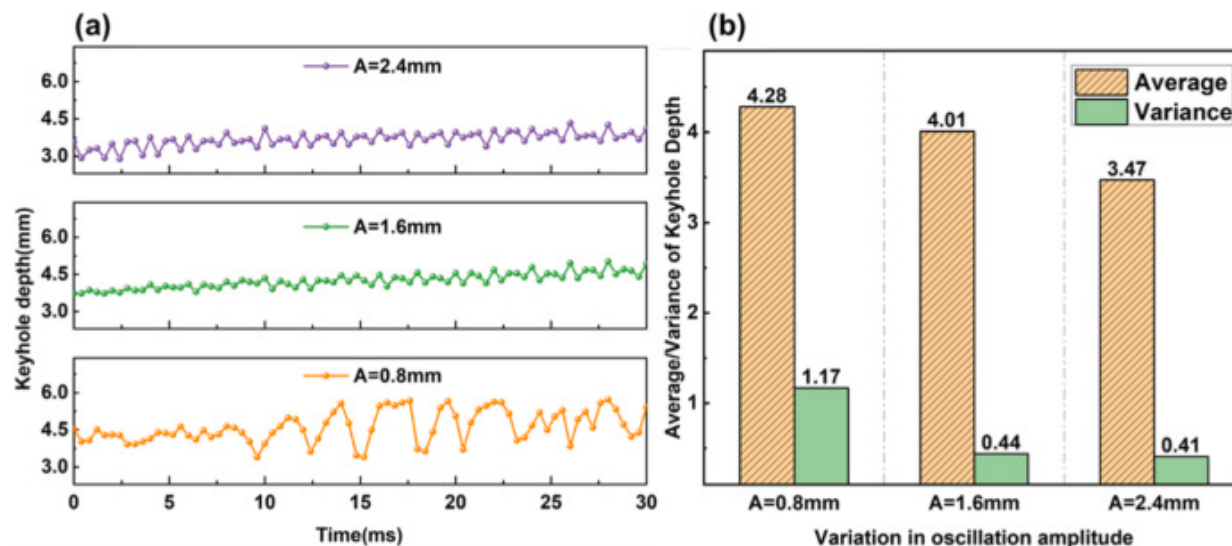
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Fig. 6. Melt flow patterns of narrow gap laser wire filling welding with circular oscillation under different amplitudes ($f=100\text{Hz}$): (a) $A=0.8\text{mm}$; (b) $A=1.6\text{mm}$; (c) $A=2.4\text{mm}$.

As the oscillation amplitude increases (Fig. 6(b and c)), the expanded scanning range produces a more uniform temperature distribution on the pool surface, reducing local temperature gradients and thereby weakening the concentrated Marangoni flow. The molten pool develops broader horizontal vortices, resulting in lower overall flow velocity. Meanwhile, the more uniform evaporation behavior alleviates the localized recoil-pressure loading on the keyhole walls, improving their mechanical stability and reducing collapse probability. The enlarged keyhole opening and diminished wall bulging further enhance the geometric stability of the keyhole under large-amplitude oscillation.

As the oscillation amplitude increases, the keyhole depth and its temporal fluctuations exhibit a clear evolutionary trend. Fig. 7(a) illustrates the temporal evolution of keyhole depth under different oscillation amplitudes, while Fig. 7(b) summarizes the corresponding mean depth and variance. At a small amplitude ($A=0.8\text{mm}$), the mean keyhole depth reaches 4.28mm with a large variance of 1.17mm , indicating pronounced and irregular fluctuations. Such behavior reflects frequent local collapse–reformation events along the keyhole wall, suggesting that the backward vortex behind the keyhole remains insufficiently suppressed and the keyhole is still governed by a distinctly non-stationary flow regime. When the amplitude is increased to 1.6mm , the mean depth slightly decreases to 4.01mm , while the variance sharply converges to 0.44mm . This reduction in fluctuation intensity implies that the oscillating beam has induced a more stable horizontal vortex within the molten pool, thereby homogenizing the forces acting on the keyhole wall and reducing transient perturbations. Further increasing the amplitude to 2.4mm results in a shallower mean depth of 3.47mm , yet the variance remains low at 0.41mm , and the depth profile becomes smoother and more closely synchronized with the oscillation frequency. This behavior indicates a transition from disturbance-dominated dynamics to a controlled response driven by forced stirring, leading to enhanced keyhole stability and reduced sensitivity to external fluctuations. Overall, as the oscillation amplitude increases from 0.8mm to 2.4mm , the mean keyhole depth decreases while the variance rapidly contracts from 1.17mm to approximately 0.4mm , demonstrating that larger oscillation amplitudes, despite lowering the energy density, substantially improve the dynamical stability of the keyhole and contribute to a more robust welding process. Consistent with the present findings, Cai et al. [19] also observed that high oscillation amplitudes led to reduced keyhole-depth fluctuation and a smoother fluctuation profile, reflecting a marked enhancement in keyhole stability.



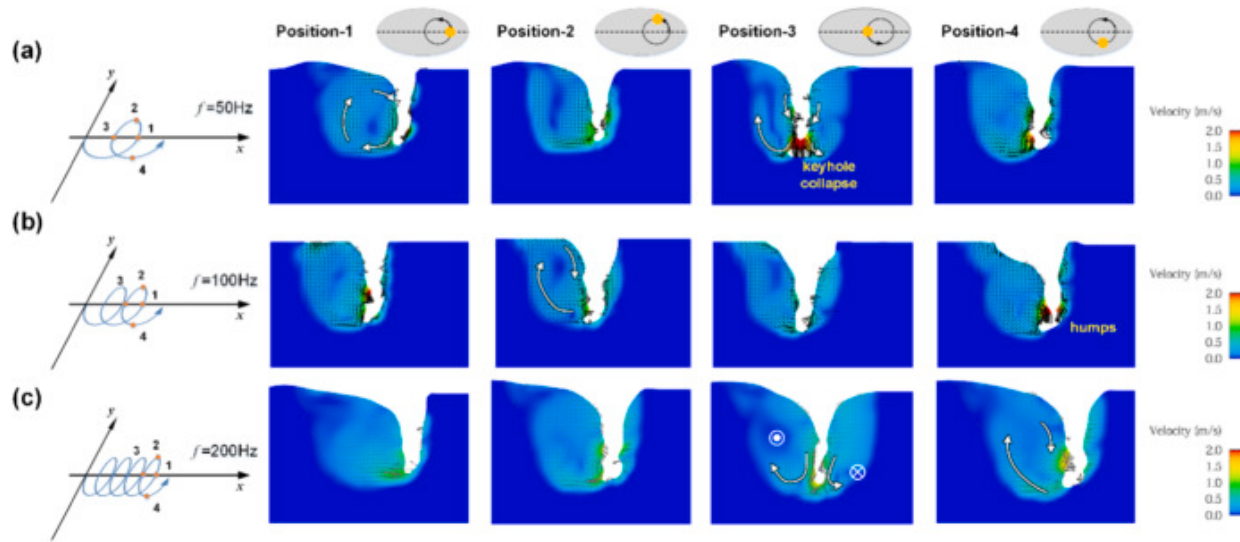
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Fig. 7. Effect of oscillation amplitude on keyhole formation in circular oscillation narrow gap laser wire filling welding: (a) Keyhole depth fluctuation; (b) Average and variance of keyhole depth.

3.1.3. Effects of frequency of melt pool and keyhole

The effects of frequency on melt flow behaviors were investigated at an oscillation amplitude of 2.4mm. Fig. 8 shows the longitudinal flow field of the melt pool at different frequencies. Similar to the influence of amplitude, increasing the frequency results in a smoother keyhole wall surface, a lower tendency for keyhole collapse, a slightly larger keyhole opening, and significantly improved keyhole stability, while the maximum melt pool flow rate decreases by 49.6%. This improvement is primarily attributed to a more uniform temperature distribution along the keyhole wall at higher frequencies, which reduces local surface tension gradients and moderates Marangoni-driven flow. In addition, more uniform vaporization mitigates transient peaks in recoil pressure, stabilizing the keyhole wall structure.



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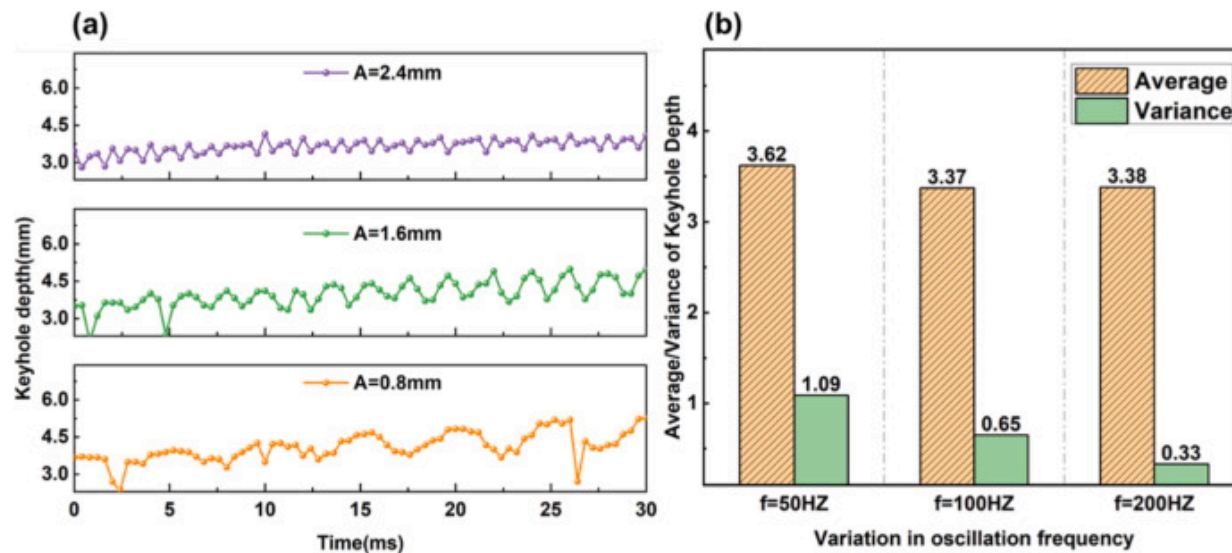
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Fig. 8. Melt flow patterns of narrow gap laser wire filling welding with circular oscillation under different frequencies: (a) $f=50\text{Hz}$; (b) $f=100\text{Hz}$; (c) $f=200\text{Hz}$.

As the frequency further increases to 200Hz , the melt flow along the keyhole wall shifts toward the bulk melt, helping maintain mechanical equilibrium of the wall and further reducing the risk of collapse. The overall flow pattern in the melt pool also changes: the vortex in front of the keyhole gradually disappears, the flow velocity behind the keyhole becomes more uniform, and longitudinal flow is enhanced, facilitating bubble escape. These observations indicate that high-frequency oscillation promotes a balance between Marangoni convection and recoil-pressure dynamics, reduces strong local disturbances caused by surface tension gradients, and generates a more controlled and stable melt pool flow, enhancing keyhole formation.

[Fig. 9\(a\)](#) illustrates the temporal evolution of keyhole depth under different oscillation frequencies, while [Fig. 9\(b\)](#) summarizes the corresponding mean depth and variance. Numerical results show that the mean keyhole depth changes only marginally with

oscillation frequency, measured at 3.62 mm, 3.37 mm, and 3.38 mm, indicating that the energy absorbed by the keyhole wall per unit time is only minimally affected by frequency. Thus, the keyhole depth is primarily governed by the oscillation amplitude, and the time required for keyhole initiation and stabilization remains comparable across different frequencies. In contrast, the fluctuation characteristics exhibit a much stronger frequency dependence. At 50 Hz, the depth variance reaches 1.09 mm, reflecting pronounced temporal instability associated with frequent local collapses. When the frequency increases to 100 Hz, the variance decreases markedly to 0.65 mm, suggesting improved coupling between the oscillating beam and molten-pool flow and a reduction in transient disturbances acting on the keyhole wall. At 200 Hz, the variance further decreases to 0.33 mm, representing a 69.7% reduction compared with the 50 Hz case, accompanied by a smoother depth profile that closely synchronizes with the oscillation frequency. High-speed imaging confirms that the enhanced stability at higher frequencies results from a substantial reduction in keyhole collapse events and a more uniform energy distribution along the keyhole wall. Overall, although adjusting the oscillation frequency can enhance process stability, both simulation and experimental results indicate that this improvement remains limited when the oscillation amplitude does not exceed a critical threshold [20]; under such conditions, increasing the frequency alone has only a modest effect on keyhole stability.



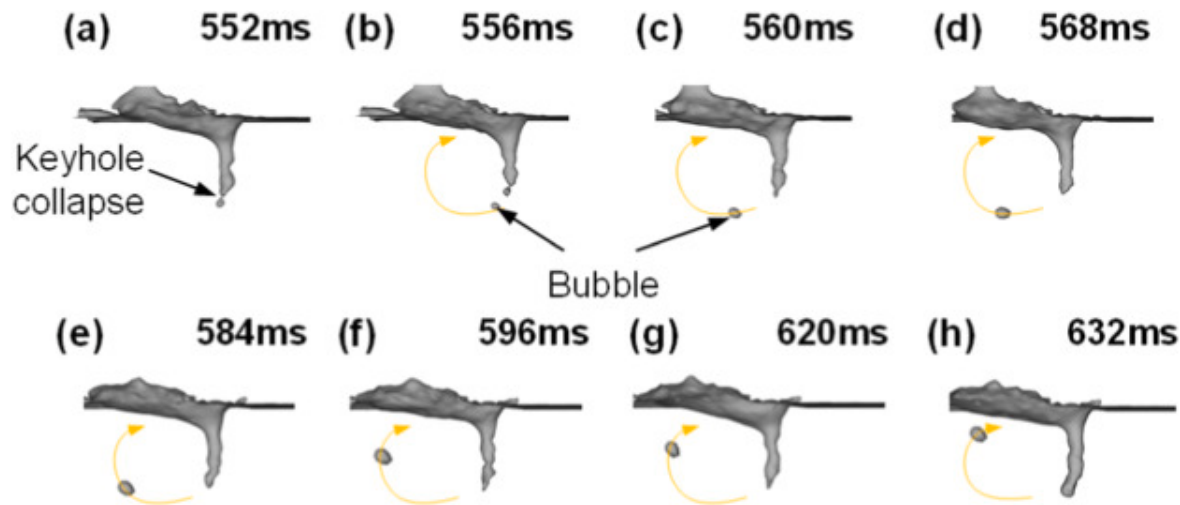
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Fig. 9. Effect of oscillation frequency on keyhole formation in circular oscillation narrow gap laser wire filling welding: (a) Keyhole depth fluctuation; (b) Average and variance of keyhole depth.

3.2. Mechanism of pore formation during laser wire filling welding

Fig. 10 shows process-induced pore generation during non-oscillating laser welding under a narrow gap groove. According to Fig. 10(a), the laser acting on the keyhole generates a considerable amount of metal vapor, leading to a mechanical imbalance on the keyhole wall surface [21]. As a result, the wall surface can barely maintain a stable morphology, causing keyhole contraction and localized bulging. Under subsequent laser-induced thermal effects and mechanical forces, the bulging zone migrates toward the keyhole bottom. As the bulge reaches the keyhole bottom, direct laser irradiation on this contracted, bulging zone easily induces localized keyhole closure, trapping the zone beneath the bulge and forming a bubble. This bubble then moves with the flow in the molten metal.



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Fig. 10. Porosity formation in narrow gap laser wire filling welding: (a) $t=552\text{ms}$; (b) $t=556\text{ms}$; (c) $t=560\text{ms}$; (d) $t=568\text{ms}$; (e) $t=584\text{ms}$; (f) $t=596\text{ms}$; (g) $t=620\text{ms}$; (h) $t=632\text{ms}$.

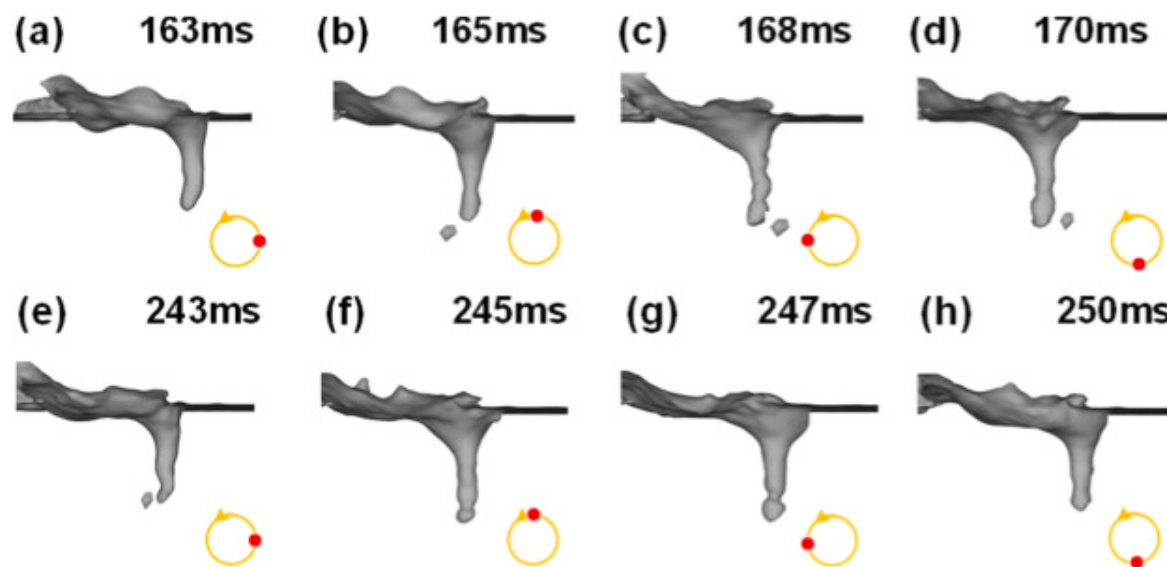
Fig. 10(b–e) shows the trajectory of bubbles in the melt pool. As observed, the bubble moves in the same direction as the overall flow field of the melt pool after detaching from the keyhole bottom. Specifically, the bubble migrates toward the tail of the melt pool. During this process, the bubble volume varies due to the complex variation of the temperature field in the melt pool and the continuously varying fluid forces acting on the bubble surface. Notably, the bubble fails to escape the melt pool when it migrates to the tail of the melt pool. This can be attributed to the presence of the flow field. According to Fig. 10(g and h), the bubble moves toward the forefront of the melt pool after approaching the surface of the molten pool. This is primarily attributed to the vortex generated behind the keyhole. Also, this phenomenon indicates that the characteristics of melt flow in non-oscillating laser welding are unfavorable for bubble escape. During the subsequent rapid

solidification of the molten pool, the trapped bubbles are more likely to form pores, thereby compromising welding quality [22].

3.3. Mechanism of pore inhibition by circular oscillating laser wire filling welding

3.3.1. Effects of flow behaviors of keyhole and melt pool

Fig. 11 shows the keyhole motion in narrow gap laser wire filling welding with circular oscillation ($A=2.4\text{mm}$, $f=100\text{Hz}$). Under high amplitudes and frequency, the keyhole showed high stability, the keyhole exhibited minor wall bulging and reduced fluctuations in keyhole length, indicating significantly decreased probability of keyhole contraction/collapse and subsequent bubble formation.



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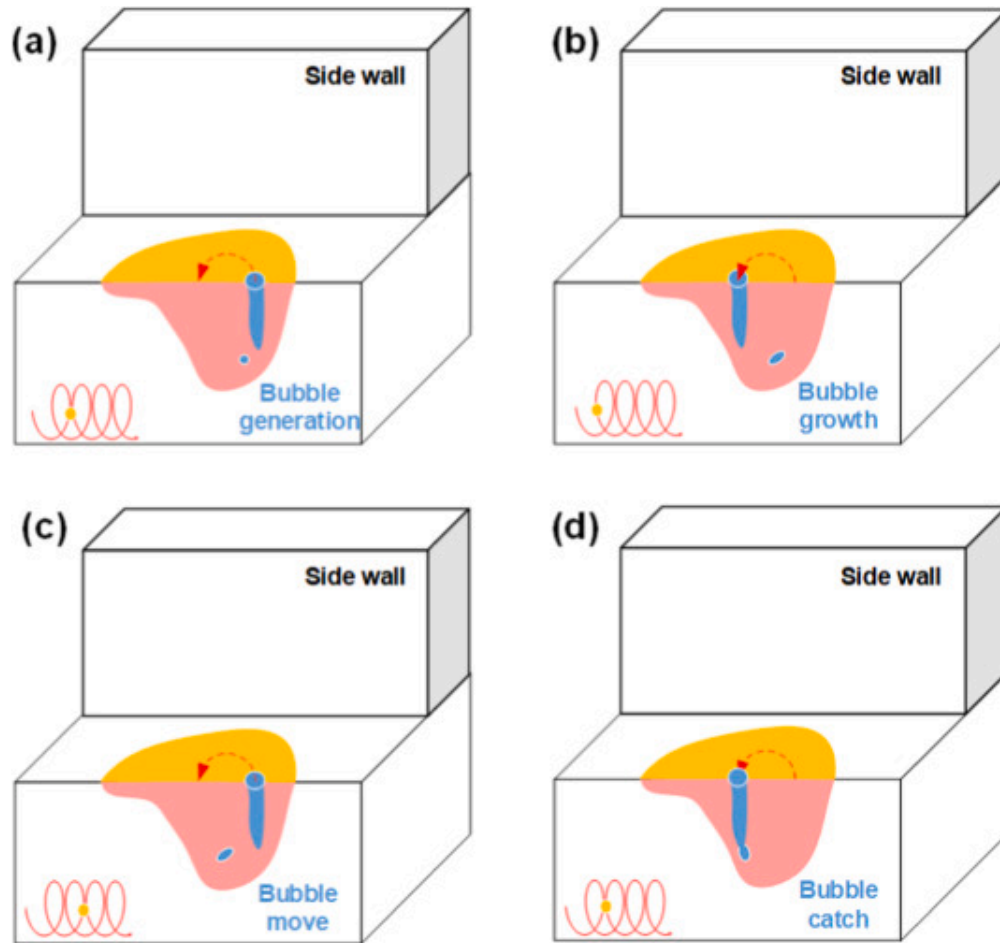
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Fig. 11. Bubble entrapment in narrow gap laser wire filling welding with circular oscillation: (a) $t=163\text{ms}$; (b) $t=165\text{ms}$; (c) $t=168\text{ms}$; (d) $t=170\text{ms}$; (e) $t=243\text{ms}$; (f)

t=245ms; (g) t=247ms; (h) t=250ms.

Notably, when a bubble is generated due to keyhole collapse at the bottom, the bubble remains trapped behind the keyhole (Fig. 11(b–f)). As the keyhole moved periodically to the rearward position of its circular trajectory, both amplitude and frequency are high, and the keyhole would encounter and recapture, and send the previously formed bubble back to the keyhole. When the keyhole moves to the next position, the original bubble in the melt pool disappears, demonstrating that the trajectory of the keyhole would cover the previous bubble formation site and recapture the bubble if the laser oscillation parameters (especially amplitude and frequency) exceed the thresholds, thereby eliminating the bubbles.

Fig. 12 shows the underlying mechanism. If the amplitude and frequency are sufficiently high, the circular motion of the keyhole in the melt pool enables the recapture of the bubble to the keyhole, allowing the entrapped gas to escape via the keyhole channel, effectively preventing pore formation [23].



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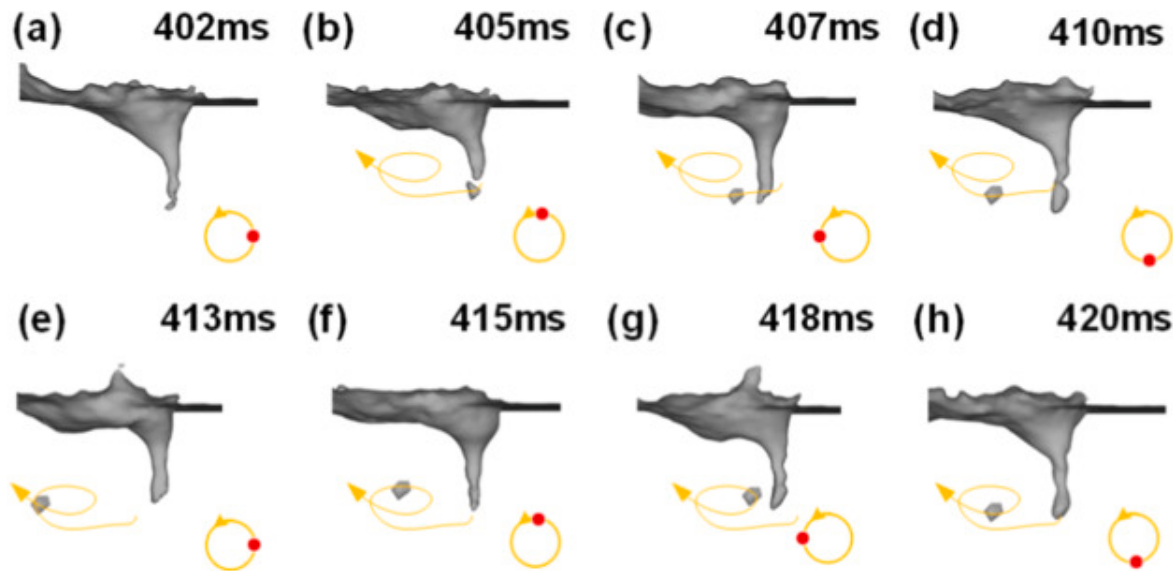
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Fig. 12. Bubble entrapment mechanism of narrow gap laser wire filling welding with circular oscillation: (a) Bubble generation; (b) Bubble growth; (c) Bubble move; (d) Bubble catch.

Overall, bubble entrapment by circular oscillating laser is the dominant cause of reduced weld porosity in narrow gap welding. Hence, the effects of oscillation parameters on bubble entrapment shall be clarified.

3.3.2. Effects of oscillation parameters

At low amplitudes ($A < 1.2$ mm, with $A = 0.4$ mm and $f = 100$ Hz as an example (Fig. 13)), the keyhole wall surface showed severe bulging (Fig. 13(a)) and its morphology was consistent with that in non-oscillating laser welding (Fig. 10), suggesting that this process is unstable. The wall surface tends to shrink inward and collapse if the curvature of its protruding part is too large, thereby forming bubbles (Fig. 13(b)). In this case, the range of keyhole oscillation is extremely small (Fig. 13(c)), and the keyhole can barely affect the bubble at the keyhole bottom even if it moves to the rearward position of the oscillation circle. Eventually, the bubble is captured by the solidification front, and a pore is generated. Fig. 13(e–h) shows the dynamic characteristics of the keyhole and the bubble in the next period after bubble formation. Low-amplitude oscillation failed to stabilize the keyhole: the keyhole length showed minor variation, while significant wall bulging was observed on the keyhole surface during the entire process [24]. Additionally, the keyhole stability varies with position: the keyhole significantly contracts and tends to collapse when it is located at the front and rear positions of the circular trajectory (Fig. 13(e–g)); the keyhole stability is slightly improved when it moves to the left and right sides (Fig. 13(f–h)).



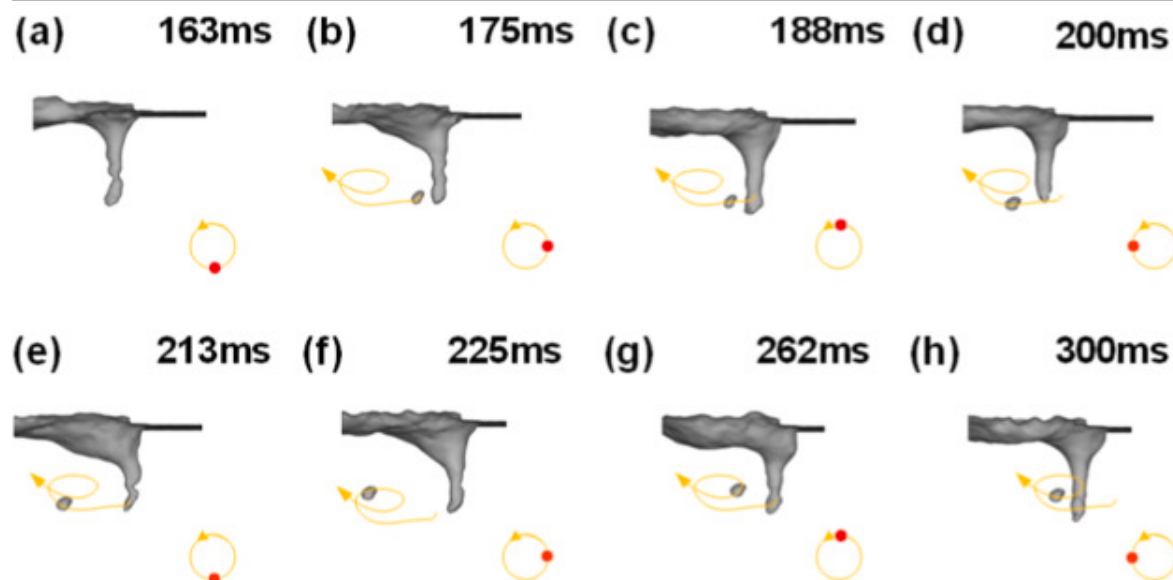
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Fig. 13. Pore generation in narrow gap laser wire filling welding with circular oscillation under low amplitude ($A=1.2\text{ mm}$, $f=100\text{ Hz}$): (a) $t=402\text{ ms}$; (b) $t=405\text{ ms}$; (c) $t=407\text{ ms}$; (d) $t=410\text{ ms}$; (e) $t=413\text{ ms}$; (f) $t=415\text{ ms}$; (g) $t=418\text{ ms}$; (h) $t=420\text{ ms}$.

Fig. 14 shows the dynamic behaviors of the keyhole at high amplitudes ($A \geq 1.6\text{ mm}$) and low frequencies ($f \leq 100\text{ Hz}$). As indicated, the increasing amplitude leads to regular fluctuations of the keyhole length during one oscillation cycle. When the keyhole is located at the front and rear positions of the trajectory, the keyhole length decreases drastically, while the wall surface of the keyhole becomes smoother (negligible bulging or wrinkling), indicating high keyhole stability in this case. When the keyhole moves to the left and right sides of the circular trajectory, the keyhole stability decreases, and the keyhole length increases. This can be attributed to the fact that the keyhole velocity was maximized as it is located on the left and right sides, leading to pronounced wall bulging and increased tendency of keyhole collapse and bubble formation [25]. When the keyhole is located at the front and rear positions, the keyhole velocity approaches the average

level, resulting in a smooth wall surface of the keyhole. This also suggests that the average keyhole velocity can serve as an indicator of the keyhole stability, as the stability is significantly improved when its velocity remains close to the average level.



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Fig. 14. Pore generation in narrow gap laser wire filling welding with circular oscillation under high amplitude and low frequency ($A=2.8\text{ mm}$, $f=20\text{ Hz}$): (a) $t=163\text{ ms}$; (b) $t=175\text{ ms}$; (c) $t=188\text{ ms}$; (d) $t=200\text{ ms}$; (e) $t=213\text{ ms}$; (f) $t=225\text{ ms}$; (g) $t=262\text{ ms}$; (h) $t=300\text{ ms}$.

According to Fig. 14(a–d), the bubble is typically generated due to keyhole collapse when the keyhole moves to the left or right sides of the trajectory. At high amplitudes, the keyhole approaches the previously generated bubble during circular motion (see Fig. 14(d)). At low frequencies, however, the keyhole moves slowly. Once it reaches the rear position, the melt pool has already moved by a considerable distance, leaving the bubbles behind without being recaptured by the keyhole. As a result, a pore is generated. Notably, the enhanced stirring effect of the keyhole motion on the melt pool under high-

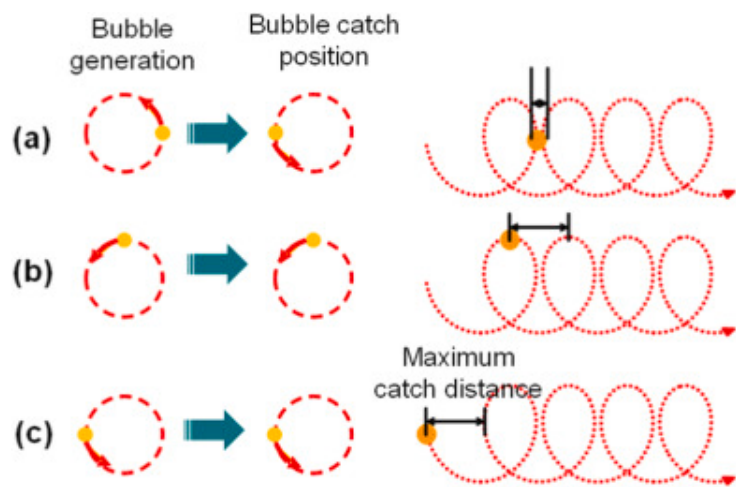
amplitude oscillation alters the bubble trajectory. Unlike conventional laser welding, the bubble follows a spiral path toward the tail of the melt pool instead of moving backward unidirectionally [26], as shown in Fig. 14 (h). As a result, the length of the bubble escape path increases significantly, making them more susceptible to entrapment in the weld by solidification, thus increasing the probability of pore formation.

Overall, the circular oscillating laser can only ensure overlapping of keyhole trajectories and achieve bubble entrapment at high amplitudes and high frequencies. In other words, oscillation parameter thresholds are present for pore elimination [27]. Equations of these thresholds were further explored.

3.4. Parameter thresholds for pore elimination by circular oscillating laser

High-frequency oscillation enables the laser beam to scan the molten region periodically, preventing continuous local energy accumulation and thereby suppressing sharp fluctuations in keyhole wall temperature and metal vapor recoil pressure. This stabilization helps maintain the system in a lower-energy dynamic steady state and reduces the likelihood of bubble nucleation. The change in oscillation frequency also reshapes the temperature gradient within the melt pool and alters the distribution of the Marangoni force. A more uniform energy input at higher frequencies weakens the transverse vortices that tend to draw bubbles into the deeper regions of the melt pool, while strengthening the longitudinal flow that guides bubbles toward the escape pathway. Meanwhile, the enlarged melt pool and reduced local solidification rate associated with higher-frequency oscillation extend the time window available for bubble migration and escape, increasing the probability of successful release before the solidification front closes. Building on these combined thermodynamic and flow effects, the subsequent analysis examines the bubble-capture distance, variations in melt-pool bottom flow velocity, and the influence of oscillation parameters on the threshold conditions for pore suppression.

Fig. 15 shows different cases of bubble entrapment by the keyhole in circular oscillating laser welding. As indicated, a maximum entrapment distance (welding direction) is present between the bubble formation site and the theoretically capturable position [28]. Therefore, if the trajectory of the circular oscillating laser (the keyhole trajectory) can effectively capture bubbles even at the maximum distance, it can be concluded that this oscillation mode completely suppresses pore formation in Al alloy welding.



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Fig. 15. Absorption distance of bubbles generated at different positions: (a) Forward generation and rear entrapment; (b) Edge generation and edge entrapment; (c) Rear generation and rear entrapment.

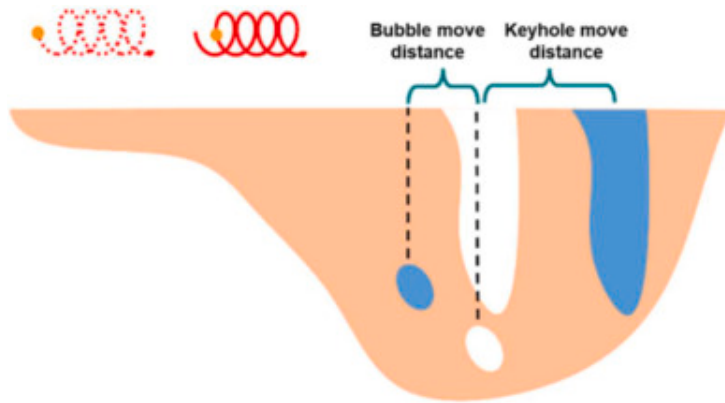
The maximum bubble entrapment distance ($L_{\max}$) was investigated. As shown in Fig. 16, the bubble entrapment distance comprises the movement distance of the generated bubble from its initial position (L_b) and the oscillation-induced movement distance of the keyhole (L_k). The movement distance of the bubble is related to the component of the flow velocity at the keyhole bottom along the welding direction. When a bubble is generated as the keyhole moves to the rearward position of the circular trajectory, the moment of bubble entrapment is defined as the moment when the keyhole reaches the

rearward position again in the next cycle (i.e., one circular oscillation period). Hence, the maximum bubble entrapment distance can be determined by:

$$L_k = V_w \times \frac{1}{f} \quad (5)$$

$$L_b = \int_{t_0}^{t_0+T} V_b, x(t) dt \quad (6)$$

$$L_{max} = L_k + L_b \quad (7)$$



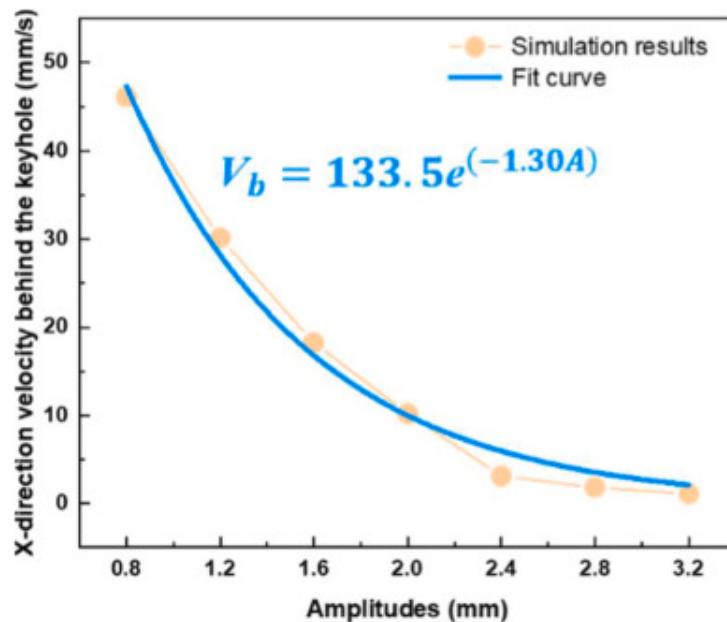
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Fig. 16. Maximum bubble entrapment distance.

where $V_b, x(t)$ is the instantaneous velocity of the bubble, which can be determined according to the component of flow velocity at the bottom of the melt pool along the welding direction. Simulation results revealed that the flow velocity at the bottom of the melt pool is significantly affected by the amplitude. Fig. 17 shows the component of flow velocity at the bottom of the melt pool along the welding direction under different amplitudes. As indicated, the flow rate is closely correlated with the amplitude. Statistical fitting indicated that flow rate can be a function of amplitude:

$$V_b = 133.5e^{(-1.30A)} \quad (8)$$



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Fig. 17. X-direction velocity at the keyhole bottom of narrow gap laser wire filling welding with circular oscillation.

Bubble entrapment primarily occurs when the keyhole trajectory passes the location of a bubble, allowing the gas in the bubble to escape from the melt pool via the keyhole.

Hence, the condition for bubble entrapment can be defined as the contact of the wall surface of the keyhole and the wall surface of the bubble, which facilitates bubble escape. The bubble entrapment distance shall meet:

$$L_k + L_b \leq \frac{d}{2} \quad (9)$$

Where d is the bottom diameter of the keyhole.

The frequency threshold for pore elimination can be determined by:

$$f \geq \frac{2 \times (V_w + 133.5e^{(-1.30A)})}{d} \quad (10)$$

Besides amplitude, the frequency threshold is also affected by keyhole diameter and welding rate, while keyhole diameter was affected by laser power, fiber diameter, welding process, and physical parameters of the material [29]. According to the simulation results, a stable bottom diameter of the keyhole was employed since the diameter of the keyhole bottom was small. Table 3 summarizes the keyhole bottom diameters under different oscillation parameters based on the simulation results. As observed, the keyhole bottom diameter was more significantly affected by the amplitude compared by the frequency. Therefore, it can be fitted as a linear function of amplitude:

$$d = 0.4 + 0.2A \quad (11)$$

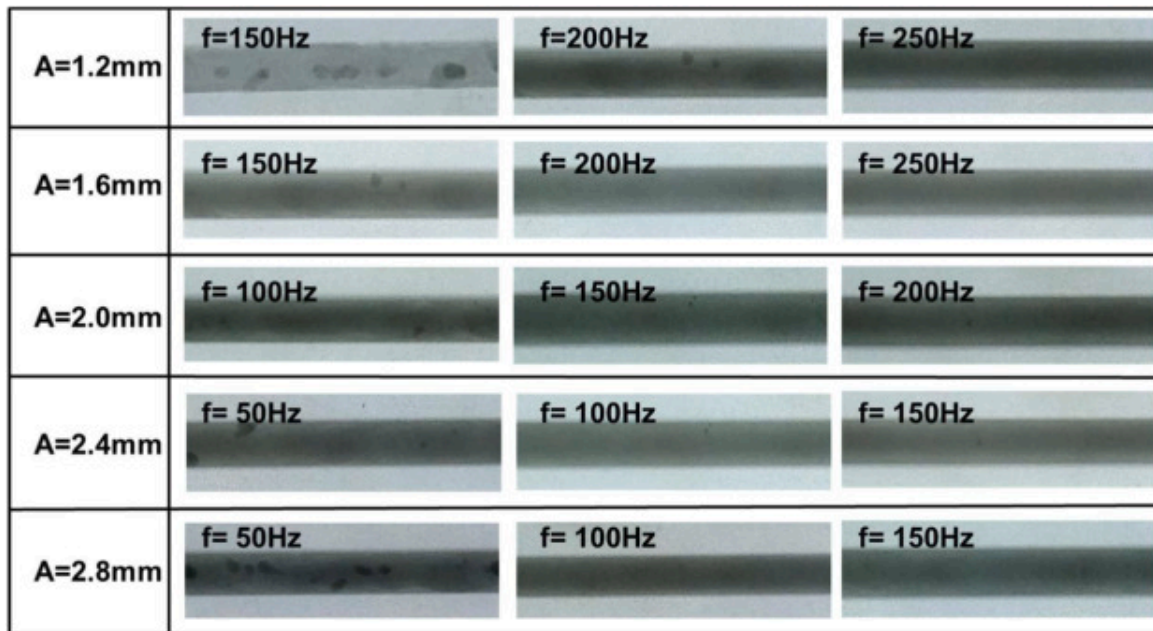
Table 3. Average size and fluctuation of keyhole bottom diameter of narrow gap laser wire filling welding with circular oscillation under different oscillation parameters.

		Bottom keyhole diameter (mm)	
		Average	Fluctuation
A	0mm	0.51	0.41
	0.8mm	0.59	0.32
	1.6mm	0.76	0.23

		Bottom keyhole diameter (mm)	
		Average	Fluctuation
<i>f</i>	2.4mm	0.87	0.11
	0Hz	0.51	0.47
	50Hz	0.79	0.24
	100Hz	0.86	0.18
	200Hz	0.91	0.14

Fig. 18 presents the results of radiographic inspection (X-ray) on welds under varying oscillation amplitudes and frequencies. Porosity, defined as the ratio of the total projected area of pores to the cross-sectional area of the entire weld bead, is used to characterize the distribution of pores within the weld. Fig. 19(a) shows the porosity distributions corresponding to different parameters obtained using XRD under different amplitudes and frequencies obtained to validate the reliability calculation. Fig. 19(b) shows the calculation results obtained by substituting keyhole diameter and welding rate into Eqs. (5), (6). The oscillation threshold of pore-free narrow gap oscillating laser wire filling welding of Al alloy can be calculated by:

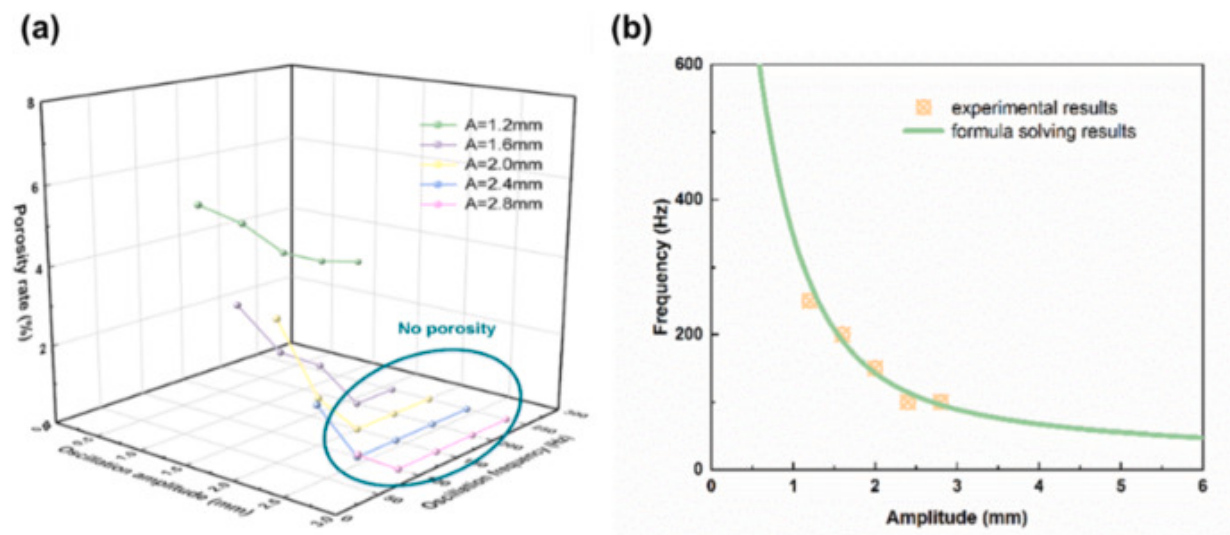
$$f \geq \frac{33.3+267e^{-1.3A}}{0.4+0.2A} \quad (12)$$



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Fig. 18. Distribution of pore defects at different oscillation frequencies and amplitudes.



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Fig. 19. Pore volume distribution and simulation results: (a) Pore volume distribution; (b) Simulation results of frequency threshold for pore inhibition.

As indicated, simulation and experimental results were consistent. As the amplitude increases, the frequency threshold corresponding to bubble elimination and keyhole stability decreases. Meanwhile, the pore-free oscillation parameters obtained by experiments were slightly lower than those obtained by calculations, further demonstrating that the proposed equation shows a high confidence level in guiding the determination of oscillating laser parameters for pore-free narrow gap filling welding of Al alloy 5A06. Building on this verification, a combined analysis of the pressure, temperature, and velocity fields further reinforces the mechanisms governing bubble transport, recapture, and keyhole stabilization. Circular oscillation reduces recoil-pressure spikes and smooths temperature gradients along the keyhole wall, suppressing collapse-induced bubble formation. At the same time, the velocity field becomes more uniform: rearward vortices weaken, while oscillation-driven longitudinal flow promotes directed

bubble migration toward the keyhole and enables periodic recapture, contributing to sustained suppression of porosity.

4. Conclusion

This chapter systematically investigates porosity suppression through numerical analysis of molten pool dynamics in narrow-gap oscillating laser wire-fed welding of aluminum alloys. The study examines the effects of oscillation patterns, amplitudes, and frequencies on porosity formation, elucidates the underlying mechanisms, and establishes threshold parameters for effective porosity elimination. Key findings include.

- (1) Circular laser oscillation significantly enhances keyhole stability by altering rear melt flow patterns, thereby reducing vortex-induced wall impacts and collapse risks. The rapid keyhole movement minimizes filler wire influence on keyhole dimensions.
- (2) While circular oscillation increases molten pool size and solidification time compared to conventional welding, the persistent rearward flow toward the keyhole impedes effective bubble escape from the pool bottom.
- (3) The porosity suppression mechanism involves dual effects: improved keyhole stability reduces bubble formation probability at the keyhole tip, while sufficient oscillation amplitude and frequency enable active bubble recapture into the rotating keyhole, providing direct escape channels and significantly lowering porosity rates.
- (4) Based on the suppression mechanism, a computational method for determining maximum bubble capture distance is developed, enabling the derivation of amplitude-dependent frequency thresholds for complete porosity elimination.

The study provides fundamental insights into process optimization for defect-free narrow-gap laser welding applications.

CRediT authorship contribution statement

Chen Liu: Writing – original draft, Methodology, Data curation, Conceptualization.

Yuanze Ma: Software, Methodology. **Pengfei Guo:** Software, Data curation. **Haoyue Li:** Methodology. **Yingke Wang:** Methodology. **Jianfeng Gong:** Software. **Liqun Li:** Supervision. **Changyun Li:** Writing – review & editing. **Hongbo Xia:** Writing – review & editing, Supervision, Methodology.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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